Pranjal Kumar Nandi

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Education

Bachelor of Science in **Computer Science and Engineering**, Class of 2021
Department of Computer Science and Engineering, **University of Dhaka**

Research Interest

Multimodal Learning, Multi-agent Collaboration, Computer Vision, Edge Intelligence

Publications and Patents

- **Pranjal Kumar Nandi**, Md Rejaul Islam Reaj, Sujan Sarker, Md Abdur Razzaque, Md Mamun-or-Rashid, and Palash Roy, "**Task offloading to edge cloud balancing utility and cost for energy harvesting internet of things.**" Journal of Network and Computer Applications 221 (2024): 103766, Elsevier Paper Link
- Abu Tyeb Azad, Ishita Sur Apan, Fahim Ahmed, . . . **Pranjal Kumar Nandi**, et al "**BaFCo: A Document Understanding Benchmark for Complex Bangla Form Comprehension**", European Conference on Computer Vision (ECCV), Malmo, Sweden, 2026. Paper Link
- David Lawrence Turock, Justin Mark Brockie, James Michael Kelly, ... **Pranjal Kumar Nandi**, et el. "**Artificial intelligence machine learning method and system for identifying and redacting personally identifiable information in a monitored environment using real-time sensor data**", US Patent Publication No. US12462909B2, Granted November 4, 2025

Research Experience

Research Collaborators, Center for Computational & Data Sciences (CCDS) July 2025 – Present

Supervisors: Dr. A K M Mahbubur Rahman, Dr. Md Mofijul Islam

- Collaborating on **test-time adaptation** in VLMs and **PyTutor**, an LLM-based pedagogical tool for learning Python.
- Contributed to **BaFCo**, a document-understanding benchmark for **complex Bangla form comprehension**, resulting in a paper published at **ECCV 2026**.
- Developing **streaming speech-driven 3D motion generation** models, investigating causal multimodal learning and knowledge distillation for low-latency, speech-aligned motion synthesis.

Undergraduate Research Assistant, Green Networking Research Group (GNR), University of Dhaka, Bangladesh Jan 2020 – Jun 2021

Supervisor: Dr. Md. Abdur Razzaque, Professor & Chairman, Department of CSE, University of Dhaka

- Researched **game theory, deep learning, optimization, and meta-heuristic methods** for **task offloading and resource allocation** in cloud computing, with a focus on IoT applications.
- Developed a novel algorithm using **Social Cognitive Optimization**, a meta-heuristic approach, to determine optimal **task offloading policies for energy-harvesting IoT devices**.
- Acquired in-depth knowledge of **research methodologies** and practical experience in conducting studies and **authoring scholarly papers**.

Professional Experience

Machine Learning Engineer II, Therap (BD) Ltd., Dhaka, Bangladesh Oct 2024 - Present

- Developed an **LLM-powered, retrieval-augmented generation (RAG) support chatbot** using user guides and support chat data; built automated data scraping and processing pipelines, improved response quality through **retrieval enhancements and prompt engineering**, and developed **benchmark datasets and automated evaluation pipelines**.
- Developed an **audio classification model** using spectrograms to detect sounds in medical cabins and care facilities; compared signal-transformation methods, collected real-world audio with **Raspberry Pi 5** and mini microphones, and refined preprocessing to reduce false positives.
- Implemented **real-time speech-to-text** using **OpenAI Whisper** for live audio streams; integrated microphone and Raspberry Pi capture and optimized preprocessing to reduce latency and transcription errors.

Focus areas: Audio Classification, LLM, RAG, Activity Recognition, VLM, Chatbot, Document AI, Benchmarking

Machine Learning Engineer, Therap (BD) Ltd., Dhaka, Bangladesh Oct 2022 - Sep 2024

- Worked on designing and building a **computer vision system** for **privacy-preserving remote monitoring** of patient activity.

- Researched **video-understanding and vision–language models**, evaluated their strengths and limitations for practical tasks, and produced recommendations to guide model selection and future development.
- Devised algorithms to aggregate and statistically analyze computer vision model outputs for **real-time, high-level activity data**.
- Contributed to a **video redaction solution** using **person detection, instance segmentation, and multi-object tracking** for precise redaction of individuals.
- Authored a **U.S. patent** for non-invasive monitoring of identifiable information in medical facilities.

Technologies and focus areas: Activity Recognition, Face Recognition, YOLO, FastAPI, React, Docker, Kafka, NVIDIA Jetson AGX Orin

Associate Machine Learning Engineer, Therap (BD) Ltd., Dhaka, Bangladesh Sep 2021 - Sep 2022

- Researched and tested **ML models and datasets** for activity recognition, pose estimation, segmentation, and face recognition in medical care applications.
- Enhanced **NVIDIA Jetson devices** and **DeepStream SDK** capabilities through research.

Technologies: C++, Python, NVIDIA DeepStream, PyTorch, Segmentation Models, NVIDIA Jetson NX

Current Research Projects

Streaming Speech-Driven 3D Motion Generation April 2026 - Present

- Developing a **causal multimodal generative model** for continuous **3D co-speech motion synthesis** from streaming audio, without future audio context.
- Investigating **knowledge distillation** and inference strategies to improve **motion realism and speech alignment** under streaming constraints.
- Designing **latency audits and evaluation protocols** to assess audio look-ahead, temporal alignment, and motion continuity in online generation.
- Supervisor: Dr. A K M Mahbubur Rahman, Dr. Md Mofijul Islam

Selected Projects

Task Offloading Policy in Mobile Edge Cloud | *Task offloading optimization for energy-harvesting IoT devices* 2021

- Designed and implemented a system and algorithm to optimize computation offloading decisions for IoT devices to edge cloud, enhancing user utility and system efficiency..

Dishari | *AI enabled pathfinder robot duo* 2019

- Dishari is an AI enabled pathfinder robot duo (ground,drone) that works using computer vision with Python.

Autonomous Traffic Monitoring and Congestion Control | *A traffic control system* 2018

- Proposed an IoT-based traffic management system leveraging edge computing and fuzzy neural networks for real-time monitoring, noise mapping, and optimal route generation.

Sumobot | *Sensor-based sumo robot* 2018

- Built an **Arduino-based sumo robot**, integrating **sensor feedback and motor control** for navigation and pushing opponents out of the ring.

Decoded | *A simple game* 2017

- Decoded is a single player jigsaw puzzle solving game. It was developed using Borland Graphics Interface (BGI) and C++.

Development Skill Set

Languages: Python, C, C++, JavaScript, MATLAB

Frameworks/Libraries: Pytorch, Tensorflow, Keras, FastAPI, Node.js, NVIDIA Deepstream

Databases/Misc: Docker, MySQL, PostgreSQL, SQLAlchemy, LaTeX, Linux, Git, Kafka, RabbitMQ

Edge Device: NVDIA Jetson NX, Jetson AGX Orin, Arduino, Raspberry Pi, ADLINK Neon

Organizations

Student Activities Coordinator, IEEE Computer Society, Student Branch, University of Dhaka 2020 – 2021

Co-founder, Techno Rex, Dhaka, Bangladesh 2017 – 2019

Achievements, Honours And Activities

Academic	**Secondary School Certificate Scholarship**, Education Board, Bangladesh Government	2014-16
	Junior School Certificate Scholarship, Education Board, Bangladesh Government	2012-14
Programming Contests	**Rank 7**, Inter University Programming Contest, Military Institute of Science & Technology	2019
	Rank 13, Inter University Programming Contest, Bangladesh University of Professionals	2019
	Rank 20, IUPC, State University of Bangladesh	2019
Hackathons & Others	**Champion**, Bracathon 3.0 (Micro-finance Category), BRAC, Bangladesh.	2019
	Runner Up, Line Follower Robot Race, North South University, Dhaka	2019
	Runner Up, Smart City Idea Competition, Robofest, University of Dhaka	2017
Math Olympiads	**District Champion**, Mathematics Olympiad, Bangladesh	2008

References

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Dept. of Computer Science and Engineering
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Department of Robotics and Mechatronics
University of Dhaka
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